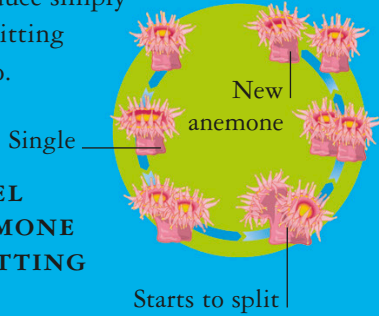




GOING IT ALONE

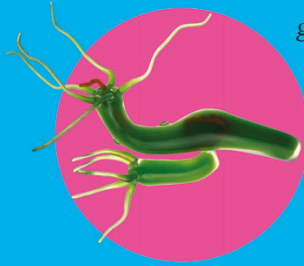
Sexual reproduction is not the only way to increase the species. Animals such as jewel anemones and flatworms can reproduce simply by splitting in two.

JEWEL ANEMONE SPLITTING



Other anemones and hydras (small water creatures) use budding, in which a new animal starts to grow as part of its mother before detaching.

HYDRA BUDDING



Parthenogenesis is another method used, particularly by aphids. Here the eggs develop without the need for fertilization by sperm. The offspring are all female, identical to their parent, and are called clones. Cloning is a useful way to increase the population quickly when conditions are good and can support extra offspring. It was thought that female Komodo dragons also do this, however recent research suggests that they actually store sperm for many months after mating.

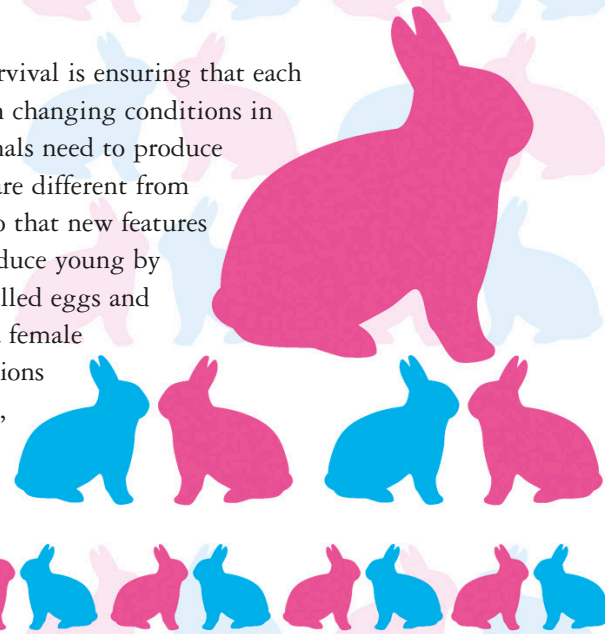
KOMODO DRAGON

When I grow up I want to be *just like you*, Mom.

The NEED

Adapt and survive

One of the biggest challenges to survival is ensuring that each generation is equipped to cope with changing conditions in their environment. To do this, animals need to produce as many offspring as they can that are different from their parents and from each other so that new features can emerge over time. Animals produce young by sexual reproduction. Special cells called eggs and sperm mix DNA from a male and a female (see p.15). That way, different variations of DNA are passed on to the young, which helps the population survive if conditions change.



Finding a mate

If you live in a large group of your own species all year round then it's not hard to find a mate, but more solitary animals or those that can't move have to find another way to produce their offspring.



Advertising—you can do this by calling, laying a scent trail, or putting on a visual display. The downside is that it may also attract unwanted attention from predators and parasites.



Lekking—the animal equivalent of going to a nightclub. Some animals, such as capercaillies or gemboks, gather at a regular breeding site where the males fight over the best spot to show off their dance moves. The females watch and rate their appearance and ability before pairing off.



Find a soul mate—many solitary animals, such as albatrosses and swans pair up for life. Even though they spend most of the year apart, they meet up again in the breeding season to mate. Pairs that do this become more successful at raising young as the years go by.